

MEMORANDUM

Date: 02/06/2026

Welcome to the team! We are flying blind right now. The shift supervisors keep blaming machine downtime for missing our daily quota, but I don't know which machines are actually the problem. I've asked IT to pull a small sample of production logs from the last 72 hours—just 55 records, located in the file `jv_production_logs.csv`.

I need you to explore this data, find out where our biggest bottlenecks are, and present a brief summary. Be careful, though IT warned me that some operators forgot to log their downtime, and there might be a typo or two.

Your Tasks in R (Please follow in order):

- **Task 1: Inspect the Data.** Load the CSV into R. Use the `summary()` function. Look closely at the `Downtime_Minutes` column. How many operators forgot to log their time?
- **Task 2: Filter for Bottlenecks.** Create a new data frame using `filter()` that only contains records where `Downtime_Minutes` is greater than 30. We only care about major stops.
- **Task 3: Prioritize by Defect.** Take the filtered data from Task 2 and use `arrange()` to sort it from highest to lowest by `Defective_Parts`. This tells us which broken machine is bleeding the most money.
- **Task 4: Group & Summarize.** Go back to your original dataset. Use `group_by()` and `summarize()` to calculate the *Total Parts Produced*, *Average Downtime*, and *Total Defective Parts* for each `Machine_Type`.
 - *Hint:* Because of the missing data you found in Task 1, R will give you an error (NA) for your average downtime unless you add `na.rm = TRUE` inside your math functions!
- **Task 5: Visualize.** Use `ggplot2` to create a Bar Chart showing the total defective parts by machine type. Look closely at the categories on the bottom of your chart—did you catch the typing mistake the operators made?

Deliverables:

1. Your documented R script (.R file) on Github.
2. The exported image of your chart (.png).
3. A 3-slide presentation answering my questions: What is the situation (mention the dirty data)? What did you find? Which machine type should I send maintenance to tomorrow morning?

Submission Date: 07/06/26